Literature Review

Hamilton, Leskovec, and Jurafsky (2016)

Hamilton et al. investigate whether changes in word meaning follow general patterns over time. They train embeddings on historical text divided into decades and compare three representations: PPMI, SVD, and skip-gram with negative sampling (SGNS). Because separately trained embeddings do not necessarily use the same coordinate system, they align them with orthogonal Procrustes before comparing vectors. Their measure of change is the cosine distance between a word's representations in two successive decades:

$$\Delta^{t}(w) = 1 - \cos(v_{w,t}, v_{w,t+1}).$$

The authors find that SGNS performs well when identifying known semantic shifts, which is one reason we chose the English SGNS vectors from HistWords. They also identify two patterns. The *law of conformity* suggests that commonly used words change more slowly; the *law of innovation* suggests that words with more meanings tend to change faster, even after accounting for frequency. They examine these relationships with a mixed-effects model:

$$\Delta^{t}(w) = \beta_{f} \log f^{t}(w) + \beta_{d} \log d^{t}(w) + \beta_{t} + z_{w} + \varepsilon.$$

Here *f* denotes frequency, *d* approximates polysemy, and *z* is a word-specific effect. Our project uses the decade-to-decade cosine distances as values to forecast rather than treating them only as evidence of past shifts. The model also suggests possible later predictors, although we have not yet added frequency or polysemy to our prepared data. One useful distinction is that the paper analyzes which factors are associated with measured change, while we are asking whether a future decade's change score can be estimated using information from earlier decades.

Schlechtweg et al. (2020)

Schlechtweg et al. present SemEval-2020 Task 1, a shared benchmark for measuring lexical semantic change. Instead of evaluating systems only through examples selected by researchers, the task uses human judgments of word usage. For English, it provides text from 1810–1860 and 1960–2010 and a set of 37 target words. Annotators compare pairs of uses and rate how closely their meanings are related. These judgments support two evaluation tasks: a binary decision about whether a sense was gained or lost, and a graded ranking of how much each word changed.

For the graded task, the reference scores are based on differences in sense distributions, measured using Jensen–Shannon distance, and systems are evaluated by Spearman rank correlation. Strong submissions often rely on static embeddings with alignment and cosine distance. This makes SemEval useful for checking whether our embedding-based distances correspond to human judgments. There is an important difference in time scale, though: SemEval compares two broad periods about a century apart, while our main data consists of consecutive decades. We would therefore need to compute a comparable long-term change measure before evaluating against its labels. SemEval is not used to train our forecasting models. We will also need to account for its part-of-speech tags, since the HistWords vocabulary does not distinguish a noun from a verb with the same spelling. This matters for targets that have multiple grammatical uses.

Dubossarsky, Grossman, and Weinshall (2017)

Dubossarsky et al. question how much of the measured movement in historical embeddings is caused by changes in meaning, rather than differences in sampling or corpus composition. They construct control datasets in which text is shuffled across decades or sampled repeatedly from one period. These controls should contain little genuine chronological change, but the resulting embeddings still show substantial variation. In particular, the apparent relationship between word frequency and semantic change remains strong even in shuffled data.

This is an important qualification to Hamilton et al.'s findings. A rare word can receive a high cosine-distance score because its embedding is estimated from fewer examples, not because its meaning has shifted. For forecasting, this means a low prediction error would not by itself demonstrate that we can anticipate linguistic change; a model might just learn which words tend to have unstable vectors. The paper motivates checking errors across frequency groups and eventually comparing our model with a baseline that captures sampling noise. For Milestone 1, it mainly affects how we should interpret the scores we have collected. The shuffled-data experiment is especially useful because it shows that recovering a correlation from historical embeddings is not enough to establish that the correlation is linguistic. We would need to show that our forecasting method beats a reasonable control, not just a baseline that always predicts zero change. This also suggests looking closely at unusually large distance values before treating them as examples of dramatic shifts in meaning.

Dataset Description

We use HistWords English SGNS embeddings for the decades from 1800 to 1990. We kept alphabetic words with valid vectors in every decade and calculated cosine distances between successive decades, leaving 12,599 words and 239,381 observed changes over 19 intervals. To prepare the forecasting data, we paired each interval's change score with the score from the preceding interval. We split by time rather than by word: 188,985 training rows for target intervals ending in 1820–1960, 12,599 development rows ending in 1970, and 25,198 test rows ending in 1980 or 1990. We also collected the SemEval-2020 English corpora, target list, and annotations for optional external evaluation. The file formats and data examples are described in *data.md*. Our current model-ready rows include the word, the target decade interval, the observed change from the previous interval, and the target interval's observed change. No model has been trained yet; these files are intended to support the baselines and evaluation in the next milestone. We retained SemEval separately because its period-level annotations measure a different outcome from our decade-by-decade targets.

References

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